Query Analysis

Query 1: Display All Student Records

SQL Query

SELECT * FROM Students;

Result

Displays all student records stored in the **Students** table.

Explanation

This query retrieves every record from the Students table. It is mainly used to verify that all student information has been inserted correctly into the database.

StudentID	Name	Gender	Age	Grade	MathsScore	ScienceScore	EnglishScore
1	Vishva	F	20	A	85	70	90
2	Meet	M	20	B	75	80	90
3	Saxi	F	19	C	65	78	92
4	Yashvi	F	18	A	84	75	80
5	Neha	F	21	A	80	75	91
6	Rahul	M	20	A	84	80	60
7	Arjun	M	19	C	45	80	90
8	Kavya	F	19	B	75	70	50
9	Pooja	F	20	A	75	79	97
10	Rohan	M	18	A	86	70	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Query 2: Average Score in Each Subject

SQL Query

SELECT
ROUND(AVG(MathsScore),2) AS Avg_MathsScore,
ROUND(AVG(ScienceScore),2) AS Avg_ScienceScore,
ROUND(AVG(EnglishScore),2) AS Avg_EnglishScore
FROM Students;

Result

Returns the average marks obtained by students in Mathematics, Science, and English.

Explanation

This query uses the **AVG()** aggregate function to calculate the average score for each subject. It helps evaluate the overall academic performance of the class.

AverageMathsScore
75.4000

Query 3: Top Performer (Highest Total Score)

SQL Query

```
SELECT StudentName,
(MathsScore + ScienceScore + EnglishScore) AS Total_Score
FROM Students
ORDER BY Total_Score DESC
LIMIT 1;
```

Result

Displays the student who achieved the highest total score.

Explanation

The total marks are calculated by adding the scores of all three subjects. The results are sorted in descending order, and the student with the highest total score is displayed.

StudentID	Name	Gender	Age	Grade	MathsScore	ScienceScore	EnglishScore	TotalMarks
9	Resets all sorted columns			A	75	79	97	251

Query 4: Count Students Per Grade

SQL Query

```
SELECT Grade,
COUNT(*) AS Total_Students
FROM Students
GROUP BY Grade;
```

Result

Displays the number of students in each grade.

Explanation

This query groups students according to their grade and counts the number of students in each group using the **COUNT()** function.

	Grade	TotalStudents
▶	A	6
	B	2
	C	2

Query 5: Average Score by Gender

SQL Query

```
SELECT Gender,
ROUND(AVG((MathsScore + ScienceScore + EnglishScore)/3.0),2) AS Avg_Score
FROM Students
GROUP BY Gender
ORDER BY Avg_Score DESC;
```

Result

Displays the average score of male and female students.

Explanation

The query calculates the average marks obtained by each gender. It helps compare the academic performance between male and female students.

	Gender	AvgMaths	AvgScience	AvgEnglish
▶	F	77.3333	74.5000	83.3333
	M	72.5000	77.5000	75.0000

Query 6: Students with Math Score Greater Than 80

SQL Query

```
SELECT StudentName, MathScore
FROM Students
WHERE MathScore > 80
ORDER BY MathScore DESC;
```

Result

Displays all students who scored more than 80 marks in Mathematics.

Explanation

This query filters student records using the **WHERE** clause and displays only those students whose Mathematics score is greater than 80.

StudentID	Name	Gender	Age	Grade	MathsScore	ScienceScore	EnglishScore
1	Vishva	F	20	A	85	70	90
4	Yashvi	F	18	A	84	75	80
6	Rahul	M	20	A	84	80	60
10	Rohan	M	18	A	86	70	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Query 7: Update Student Grade

SQL Query

UPDATE Students
SET Grade='A'
WHERE StudentName='Rahul Verma';

Verification Query

SELECT StudentName, Grade
FROM Students
WHERE StudentName='Rahul Verma';

Result

The student's grade is updated successfully and verified.

Explanation

The **UPDATE** statement modifies an existing record in the database. The verification query confirms that the student's grade has been updated correctly.